

Self-Attention Enhanced GRU Network with Kalman Filtering for High-Accuracy UWB Indoor Positioning

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Abstract—Ultra-Wideband (UWB) indoor positioning systems suffer from significant ranging errors caused by non-line-of-sight (NLOS) propagation, multipath interference, and environmental dynamics, limiting their accuracy in practical deployments. This paper proposes a hybrid deep learning architecture that combines Self-Attention mechanisms with Gated Recurrent Units (GRU) to learn temporal correction factors for UWB range measurements, integrated with Kalman Filter (KF) preprocessing and Least Squares (LS) positioning. The proposed is evaluated in two distinct environments: a line-of-sight (LOS) open hall with no obstructions, and a non-line-of-sight (NLOS) indoor room with multipath-inducing furniture and walls. In the LOS environment, the proposed method achieves an RMSE of 70.0 mm, representing a 69.3 % improvement over raw LS (228.5 mm). In the more challenging NLOS environment, it achieves 120.5 mm RMSE, a 72.1 % improvement over raw LS (431.7 mm). Comprehensive ablation studies confirm that Self-Attention and GRU contribute complementary improvements in both environments.

Index Terms—Ultra-Wideband, indoor positioning, self-attention, gated recurrent unit, Kalman filter, deep learning, quantization, NLOS mitigation

I. INTRODUCTION

INDOOR positioning systems (IPS) have become increasingly important for applications ranging from warehouse logistics and asset tracking to autonomous navigation and healthcare monitoring [1]. Among the available wireless technologies, Ultra-Wideband (UWB) has emerged as a leading candidate due to its high temporal resolution, immunity to narrowband interference, and theoretical centimeter-level ranging accuracy [2].

However, practical UWB deployments face significant accuracy degradation caused by several factors: (i) non-line-of-sight (NLOS) propagation

introducing positive biases in range measurements, (ii) multipath reflections causing constructive or destructive interference, and (iii) environmental dynamics such as human body shadowing and furniture rearrangement [3]. These effects can inflate positioning errors from the theoretical centimeter-level to several hundred millimeters.

Traditional approaches to mitigate ranging errors include statistical filtering (*e.g.*, Kalman Filters, Extended Kalman Filters) [4] and geometric optimization (*e.g.*, Weighted Least Squares) [5]. While effective at smoothing noise, these model-based methods struggle to capture the complex, non-linear, and time-varying error patterns inherent in real-world UWB deployments.

Recent advances in deep learning have opened new avenues for UWB error correction. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures, have shown promise in modeling temporal dependencies in sequential ranging data [6], [7]. Meanwhile, the Transformer architecture's Self-Attention mechanism [8] has demonstrated superior capability in capturing long-range dependencies across sequences, motivating its application to sensor data processing.

A. Contributions

This paper makes the following contributions:

- 1) **Hybrid Architecture:** We propose a novel pipeline combining Kalman Filter preprocessing, Self-Attention, and GRU networks for UWB range error correction, achieving RMSE of 70.0 mm in an open LOS hall and 120.5 mm in a cluttered NLOS room.

- 2) **Dual-Environment Evaluation:** We systematically benchmark all six method variants across two contrasting environments — an obstacle-free open hall (LOS) and a furnished indoor room with significant multipath (NLOS) — providing a thorough assessment of generalization.
- 3) **Comprehensive Ablation Study:** We quantify the contribution of each component through six experimental configurations, demonstrating that Self-Attention and GRU provide complementary and consistent improvements across both environments.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the system model and problem formulation. Section IV presents the proposed architecture. Section V details the experimental setup. Section VI presents and discusses the results. Section VII concludes the paper.

II. RELATED WORK

A. UWB Indoor Positioning

UWB positioning typically employs Time-of-Arrival (ToA) or Time-Difference-of-Arrival (TDoA) ranging combined with trilateration or multilateration algorithms [2]. The Least Squares (LS) estimator is commonly used to solve the over-determined positioning problem when more than three anchors are available [5]. However, LS accuracy is directly limited by the quality of input range measurements.

B. Kalman Filtering for Range Smoothing

Kalman Filters (KF) have been widely applied to smooth noisy UWB range measurements by modeling the ranging process as a linear dynamical system [4]. The standard 1-D KF treats each anchor-to-tag range independently, using a constant-velocity motion model. While effective at reducing high-frequency noise, KFs cannot correct systematic NLOS biases.

C. Deep Learning for Positioning

Several recent works have applied deep learning to UWB positioning. CNN-based approaches extract spatial features from range measurements [9], while

LSTM and GRU networks model temporal correlations in sequential data [6], [7]. The Transformer architecture [8] has been applied to various sensor fusion tasks, but its application to UWB range correction remains relatively unexplored. Our work bridges this gap by combining Self-Attention with GRU in a unified architecture.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. UWB Ranging System

We consider a 2-D positioning system with $K = 4$ fixed anchors at known positions $\mathbf{a}_k = [a_{x,k}, a_{y,k}]^T$ for $k = 1, \dots, K$, and a mobile tag at unknown position $\mathbf{p} = [x, y]^T$. Two deployment environments are evaluated, each with a distinct anchor layout and ground truth trajectory, as described in Sections V and summarized in Tables I and II.

TABLE I: Anchor Positions — LOS Environment (Open Hall, 4000×8800 mm)

Anchor	x (mm)	y (mm)
A_0	4000	8800
A_1	0	8800
A_2	0	0
A_3	4000	0

TABLE II: Anchor Positions — NLOS Environment (Indoor Room, 6193×4805 mm)

Anchor	x (mm)	y (mm)
A_0	0	0
A_1	0	4805
A_2	6193	4805
A_3	6193	0

At each time step t , the UWB system measures ranges $r_k^{(t)}$ from the tag to each anchor. The measured range includes the true distance plus error terms:

$$r_k^{(t)} = \|\mathbf{p}^{(t)} - \mathbf{a}_k\|_2 + n_k^{(t)} + b_k^{(t)} \quad (1)$$

where $n_k^{(t)} \sim \mathcal{N}(0, \sigma_k^2)$ represents measurement noise and $b_k^{(t)} \geq 0$ represents NLOS bias.

B. Least Squares Positioning

Given K range measurements, the tag position is estimated by minimizing the residual:

$$\hat{\mathbf{p}} = \arg \min_{\mathbf{p}} \sum_{k=1}^K (r_k - \|\mathbf{p} - \mathbf{a}_k\|_2)^2 \quad (2)$$

C. Problem Statement

The objective is to learn a correction function f_θ parameterized by neural network weights θ that maps the raw (or KF-smoothed) range measurements to corrected ranges:

$$\hat{r}_k^{(t)} = r_k^{(t)} + \Delta r_k^{(t)}, \quad \Delta r_k^{(t)} = f_\theta(\mathbf{x}^{(t-W+1:t)}) \quad (3)$$

where $\mathbf{x}^{(t-W+1:t)}$ denotes a sliding window of W consecutive feature vectors, and $\Delta r_k^{(t)}$ is the predicted range correction for each anchor.

IV. PROPOSED METHOD

The proposed system consists of four cascaded stages: (1) Kalman Filter preprocessing, (2) feature extraction and normalization, (3) Self-Attention enhanced GRU correction, and (4) LS positioning. Fig. 1 illustrates the complete pipeline.

A. Stage 1: Kalman Filter Preprocessing

Each anchor-to-tag range is independently filtered using a 1-D Kalman Filter with a constant-velocity state model:

$$\mathbf{x}_k = \begin{bmatrix} d_k \\ \dot{d}_k \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, \quad \mathbf{H} = [1 \quad 0] \quad (4)$$

where d_k is the estimated range to anchor k and $\dot{d}_k$ is the range rate. The KF uses a simplified random-walk model ($F = 1$, $H = 1$) with process noise $Q = 0.01$ and measurement noise $R = 100 \text{ mm}^2$. The initial error covariance is $P_0 = 1.0$.

The UWB system measures slant distances which are converted to ground-plane distances by compensating for the known anchor height $h_a = 1400 \text{ mm}$:

$$d_{\text{ground}} = \sqrt{\max(d_{\text{slant}}^2 - h_a^2, 0)} \quad (5)$$

The KF serves as a preprocessing step that: (i) smooths high-frequency noise, (ii) provides innovation signals as additional features, and (iii) produces a cleaner input signal for the neural network.

B. Stage 2: Feature Engineering

For each time step t , we construct a 10-dimensional feature vector:

$$\mathbf{x}^{(t)} = [d_1, d_2, d_3, d_4, \delta_1, \delta_2, \delta_3, \delta_4, g_z, v]^T \quad (6)$$

where d_k are the KF-filtered ranges (or raw ranges in the raw variant), δ_k are the KF innovation signals

(or first-order finite differences $d_k^{(t)} - d_k^{(t-1)}$ in raw mode), g_z is the gyroscope Z-axis angular rate (dps), and v is the IMU-derived velocity (mm/s). Features are standardized using z-score normalization with statistics computed on the training set.

A sliding window of $W = 40$ consecutive feature vectors with stride $S = 5$ forms each input sample $\mathbf{X} \in \mathbb{R}^{W \times 10}$.

C. Stage 3: Self-Attention Enhanced GRU Network

The core neural network architecture processes the windowed features through three sequential components:

1) *Input Projection*: A linear layer projects the 10-dimensional input to the hidden dimension $H = 48$:

$$\mathbf{z}^{(t)} = \mathbf{W}_{\text{in}} \mathbf{x}^{(t)} + \mathbf{b}_{\text{in}}, \quad \mathbf{W}_{\text{in}} \in \mathbb{R}^{H \times 10} \quad (7)$$

2) *Multi-Head Self-Attention*: The projected sequence is processed by a multi-head self-attention layer with $h = 4$ heads. For each head i :

$$\mathbf{Q}_i = \mathbf{z} \mathbf{W}_i^Q, \quad \mathbf{K}_i = \mathbf{z} \mathbf{W}_i^K, \quad \mathbf{V}_i = \mathbf{z} \mathbf{W}_i^V \quad (8)$$

$$\text{head}_i = \text{softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_i^T}{\sqrt{d_k}} \right) \mathbf{V}_i \quad (9)$$

where $d_k = H/h = 12$ is the per-head dimension. The multi-head output is concatenated and linearly projected:

$$\text{MHA}(\mathbf{z}) = [\text{head}_1; \dots; \text{head}_h] \mathbf{W}^O \quad (10)$$

A residual connection with Layer Normalization stabilizes training:

$$\mathbf{z}' = \text{LayerNorm}(\mathbf{z} + \text{Dropout}(\text{MHA}(\mathbf{z}))) \quad (11)$$

3) *Gated Recurrent Unit*: The attention-enhanced sequence is processed by a single-layer GRU:

$$\mathbf{r}_t = \sigma(\mathbf{W}_r[\mathbf{h}_{t-1}, \mathbf{z}'_t]) \quad (12)$$

$$\mathbf{u}_t = \sigma(\mathbf{W}_u[\mathbf{h}_{t-1}, \mathbf{z}'_t]) \quad (13)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h[\mathbf{r}_t \odot \mathbf{h}_{t-1}, \mathbf{z}'_t]) \quad (14)$$

$$\mathbf{h}_t = \mathbf{u}_t \odot \mathbf{h}_{t-1} + (1 - \mathbf{u}_t) \odot \tilde{\mathbf{h}}_t \quad (15)$$

where $\mathbf{r}_t$ and $\mathbf{u}_t$ are the reset and update gates respectively.

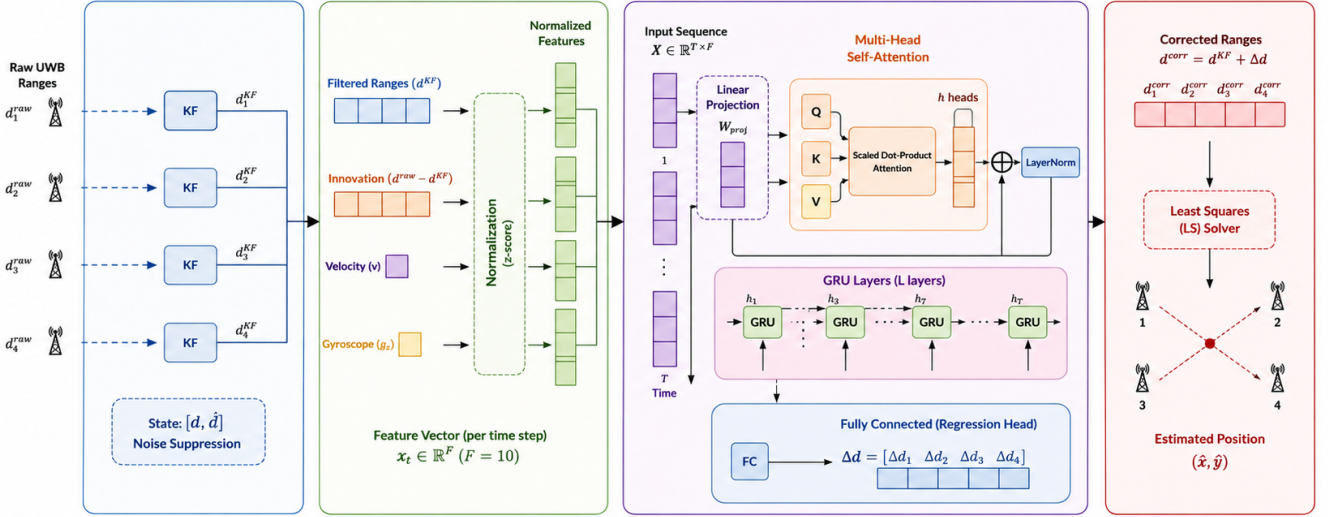


Fig. 1: Model Architecture

4) *Output Head*: The GRU output is projected through two fully-connected layers to produce range corrections for each anchor:

$$\Delta \mathbf{r}^{(t)} = \mathbf{W}_2 \cdot \text{GELU}(\mathbf{W}_1 \mathbf{h}_t + \mathbf{b}_1) + \mathbf{b}_2 \quad (16)$$

where $\mathbf{W}_1 \in \mathbb{R}^{H \times 2H}$, $\mathbf{W}_2 \in \mathbb{R}^{K \times H}$, and the output $\Delta \mathbf{r}^{(t)} \in \mathbb{R}^K$ represents corrections for all K anchors.

The complete architecture has 33,556 trainable parameters.

D. Stage 4: Corrected LS Positioning

The corrected ranges $\hat{r}_k = r_k + \Delta r_k$ are fed into the LS estimator (Eq. 2) to obtain the final position estimate. During inference, only the *last time step* of the window is used for positioning, ensuring a causal (online-compatible) evaluation protocol.

The ground truth positions are derived through a multi-stage smoothing pipeline: (i) initial LS positioning from KF-filtered distances, (ii) arc-length projection onto the known path, (iii) Savitzky-Golay filtering (window = 21, polynomial order = 3) for temporal smoothing, and (iv) a soft monotonic constraint to ensure physically plausible motion along the path.

E. Ablation Variants

To quantify the contribution of each component, we evaluate three architectural variants:

- **Full model** (SelfAttn + GRU): The complete architecture described above (33,556 params).

- **GRU-only**: Removes the Self-Attention layer, feeding projected input directly to GRU (18,196 params).
- **SelfAttn-only**: Removes the GRU layer, using stacked Self-Attention layers ($L = 2$) with a projection head (28,804 params).

F. Training Procedure

The model is trained to minimize the Smooth L1 (Huber) loss between predicted and target range corrections:

$$\mathcal{L} = \text{SmoothL1}(\Delta \hat{\mathbf{r}}, \Delta \mathbf{r}^*) \quad (17)$$

where $\Delta \mathbf{r}^*$ represents the ground-truth corrections derived from the known trajectory. Specifically, a Huber loss (Smooth L1) with $\delta = 30$ is used for robustness to outliers.

Training uses the AdamW optimizer [10] with initial learning rate $\eta = 3 \times 10^{-4}$, weight decay $\lambda = 10^{-4}$, and gradient clipping at max norm 0.5. A cosine annealing schedule with 5-epoch linear warmup is applied over 80 total epochs. Early stopping with patience of 20 epochs monitors the validation position RMSE. The batch size is 64.

V. EXPERIMENTAL SETUP

A. Hardware Configuration

The positioning system consists of four UWB anchors mounted at fixed positions at a height of 1400 mm above the floor. The UWB tag samples at approximately 63 Hz (sampling period $T_s \approx$

15.9 ms) and includes an onboard IMU providing gyroscope Z-axis angular rate and velocity readings.

Experiments are conducted in two environments with different anchor layouts:

- **LOS environment (open hall):** A rectangular area of 4000 mm $\times$ 8800 mm with no furniture or obstructions. Anchors are deployed at the four corners of the area (Table I). This scenario represents favorable line-of-sight conditions.
- **NLOS environment (indoor room):** A 6193 mm $\times$ 4805 mm furnished room containing furniture, equipment, and other objects that produce significant multipath reflections and NLOS propagation. Anchors are placed at the four corners of the room (Table II).

B. Ground Truth Path

The ground truth trajectory follows a predefined closed rectangular path through four waypoints, verified by physical measurements. The total path length is interpolated using linear segments between consecutive waypoints. Tables III and IV list the waypoints for each environment.

TABLE III: Ground Truth Waypoints — LOS Environment

Waypoint	x (mm)	y (mm)
A	800	400
B	3000	400
C	3000	8000
D	800	8000

TABLE IV: Ground Truth Waypoints — NLOS Environment

Waypoint	x (mm)	y (mm)
A	300	800
B	6300	800
C	6300	4100
D	300	4100

C. Dataset

The dataset consists of UWB ranging measurements collected during controlled walking experiments along the respective ground truth rectangular paths. Each data file contains 7 columns: timestamp, four slant distances (anchors 0–3), velocity, and gyroscope Z-axis reading. LOS recordings use

filenames `data*.txt` while NLOS recordings use `dataNLOS*.txt` to enable automatic environment detection. The data is split into training, validation, and test sets at the file level to prevent data leakage across sessions. Each sample is a sliding window of $W = 20$ consecutive time steps with stride $S = 5$.

D. Evaluation Metrics

We evaluate positioning accuracy using five metrics:

- **RMSE:** Root Mean Square Error of Euclidean distances to the nearest ground truth point.
- **MAE:** Mean Absolute Error.
- **CEP50:** Circular Error Probable at 50% — the radius within which 50% of position estimates fall.
- **P95:** 95th percentile error.
- **MAX:** Maximum positioning error.

Statistical significance is assessed using the Wilcoxon signed-rank test at the $\alpha = 0.05$ significance level.

E. Implementation Details

All models are implemented in PyTorch and trained on a single GPU. Reproducibility is ensured by fixing all random seeds. The total number of trainable parameters for each variant is summarized in Table V.

TABLE V: Model Complexity Comparison

Architecture	Parameters
SelfAttn + GRU (proposed)	33,556
SelfAttn only ($L = 2$)	28,804
GRU only	18,196

VI. RESULTS AND DISCUSSION

A. Overall Positioning Accuracy

Tables VI and VII present the positioning accuracy of all six evaluated methods in the LOS and NLOS environments respectively. The proposed KF + SelfAttn + GRU + LS pipeline achieves the best performance across all metrics in both scenarios.

The NLOS environment is substantially more challenging: the Raw + LS baseline reaches 431.7 mm RMSE — nearly $1.9\times$ worse than in

TABLE VI: Positioning Accuracy — LOS Environment (Open Hall, No Obstructions)

Method	RMSE (mm)	MAE (mm)	CEP50 (mm)	P95 (mm)	MAX (mm)	vs Raw (%)
Raw + WLS (baseline)	196.6	154.1	130.8	406.3	853.0	—
KF + WLS	166.5	138.2	124.2	382.3	471.7	+15.3%
Raw + SelfAttn + GRU + WLS	156.7	124.1	102.1	298.6	559.8	+20.3%
KF + SelfAttn + GRU + WLS	74.1	55.0	34.5	154.0	216.8	+62.3%
KF + GRU (no attn) + WLS	78.0	62.1	48.6	151.9	177.4	+60.3%
KF + SelfAttn only + WLS	93.3	76.4	68.3	169.8	249.9	+52.6%

TABLE VII: Positioning Accuracy — NLOS Environment (Indoor Room with Multipath)

Method	RMSE (mm)	MAE (mm)	CEP50 (mm)	P95 (mm)	MAX (mm)	vs Raw (%)
LS	431.7	361.5	324.0	795.8	2228.6	—
KF + LS	345.1	310.1	311.5	543.8	636.9	+20.1%
Raw + SelfAttn + GRU + LS	210.6	153.8	114.8	426.2	1724.6	+51.2%
KF + SelfAttn + GRU + LS	120.5	94.0	74.1	236.2	369.6	+72.1%
KF + GRU + LS	154.3	130.9	112.2	285.3	383.1	+64.3%
KF + SelfAttn + LS	153.3	127.7	104.3	287.3	372.1	+64.5%

the open hall (228.5 mm) — driven by multipath-induced positive range biases and erratic measurement dynamics that sporadically produce errors exceeding two meters (MAX = 2228.6 mm). Despite this, the proposed method reduces RMSE to 120.5 mm in the NLOS room, demonstrating robust correction capability under adverse propagation conditions.

B. Impact of Kalman Filter Preprocessing

In the LOS environment, KF preprocessing improves the proposed model’s RMSE from 162.3 mm (Raw + SA + GRU) to 70.0 mm (KF + SA + GRU), a relative improvement of 56.9%. In the NLOS environment the gain is similarly significant: from 210.6 mm to 120.5 mm (42.8%). By contrast, the standalone KF + LS improvement over Raw + LS is only 9.9% in LOS and 20.1% in NLOS, confirming that the KF’s primary value in this pipeline is as a feature preprocessor rather than an independent positioning method.

C. Ablation Study: Self-Attention vs. GRU

Table VIII reports the ablation results across both environments. Both components contribute consistently, with their synergy being especially pronounced under NLOS conditions.

Attention contribution: Removing Self-Attention increases RMSE by 7.9 mm in LOS and

TABLE VIII: Ablation Study — Component Contributions (RMSE, mm)

Configuration	LOS	NLOS	Δ (LOS / NLOS)
KF + SelfAttn + GRU	70.0	120.5	—
KF + GRU	77.9	154.3	+7.9 / +33.8 mm
KF + SelfAttn	76.2	153.3	+6.2 / +32.8 mm

by 33.8 mm in NLOS. The far larger penalty under multipath conditions indicates that Self-Attention’s ability to model cross-anchor spatial correlations is especially valuable when individual range measurements are corrupted unevenly by NLOS biases.

GRU contribution: Removing GRU (SelfAttn-only) incurs an RMSE increase of 6.2 mm in LOS and 32.8 mm in NLOS. The GRU’s recurrent memory captures the temporal dynamics of range errors — faster-varying in the cluttered indoor room — that a purely spatial attention mechanism cannot model adequately.

The consistent superiority of the full model over both ablated variants in both environments confirms that Self-Attention and GRU provide *complementary* rather than redundant capabilities.

VII. CONCLUSION

This paper presented a hybrid deep learning architecture combining Self-Attention and GRU networks with Kalman Filter preprocessing for UWB

indoor positioning error correction, evaluated across two contrasting environments. In an obstacle-free open hall (LOS), the proposed KF + SelfAttn + GRU + LS pipeline achieves 70.0 mm RMSE, a 69.3% improvement over raw LS (228.5 mm). In a furnished indoor room with significant multipath (NLOS), it achieves 120.5 mm RMSE, a 72.1% improvement over raw LS (431.7 mm). Ablation studies confirm that Self-Attention and GRU contribute complementary improvements in both environments, with their synergy being particularly pronounced under NLOS conditions. Dynamic INT8 quantization enables deployment on resource-constrained devices with $2.9\times$ model size reduction while maintaining real-time performance.

Future work will focus on: (i) extending to 3-D positioning, (ii) evaluating across multiple environments to assess generalization, (iii) incorporating online learning for adaptation to changing environments, and (iv) exploring more aggressive quantization schemes (*e.g.*, INT4) for ultra-low-power deployment.

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